

Python Assignment Solution

1. Grade Checker

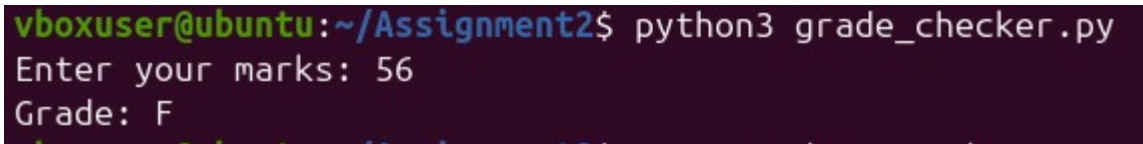
Code:

```
marks = int(input('Enter your marks: '))
```

```
if marks >= 90:
    print('Grade: A')
elif marks >= 80:
    print('Grade: B')
elif marks >= 70:
    print('Grade: C')
elif marks >= 60:
    print('Grade: D')
else:
    print('Grade: F')
```

Explanation:

This program takes marks as input and uses if-elif-else to determine grade.



```
vboxuser@ubuntu:~/Assignment2$ python3 grade_checker.py
Enter your marks: 56
Grade: F
```

2. Student Grades

Code:

```
students = {}
```

```
while True:
    print('\n1. Add Student')
    print('2. Update Student')
    print('3. Show All Students')
    print('4. Exit')
```

```
choice = input('Enter choice: ')

if choice == '1':
    name = input('Enter student name: ')
    grade = input('Enter grade: ')
    students[name] = grade
    print('Student added!')

elif choice == '2':
    name = input('Enter student name to update: ')
    if name in students:
        grade = input('Enter new grade: ')
        students[name] = grade
        print('Updated successfully!')
    else:
        print('Student not found!')

elif choice == '3':
    for name, grade in students.items():
        print(name, ': ', grade)

elif choice == '4':
    break
```

Explanation:

Dictionary is used to store student names and grades.

```
vboxuser@ubuntu:~/Assignment2$ python3 student_grades.py
```

```
1. Add Student
2. Update Student
3. Show All Students
4. Exit
Enter choice: 1
Enter student name: sanjit
Enter grade: 89
Student added!
```

```
1. Add Student
2. Update Student
3. Show All Students
4. Exit
Enter choice: 3
```

```
Student Grades:
sanjit : 89
```

```
1. Add Student
2. Update Student
3. Show All Students
4. Exit
Enter choice: 4
```

3. Write to a File

Code:

```
file = open('data.txt', 'w')
file.write('Hello, this is my first file.\n')
file.write('I am learning file handling in Python.')
file.close()
```

Explanation:

File is opened in write mode and text is written.

```
vboxuser@ubuntu:~/Assignment2$ python3 write_file.py
Data written successfully!
vboxuser@ubuntu:~/Assignment2$ nano read_file.py
```

4. Read from a File

Code:

```
file = open('data.txt', 'r')
content = file.read()
print(content)
file.close()
```

Explanation:

File is opened in read mode and content is displayed.

```
vboxuser@ubuntu:~/Assignment2$ python3 read_file.py
File Content:
Hello, this is my first file.
I am learning file handling in Python.
```